

Aprendizagem Computacional (AC) Machine Learning (ML)



Introduction (I)

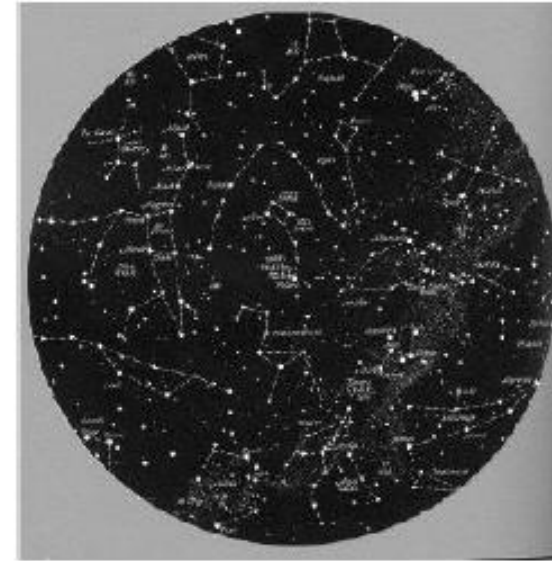
César Teixeira, DEI-FCTUC, University of Coimbra
2026-2027

ML → Recognize Patterns

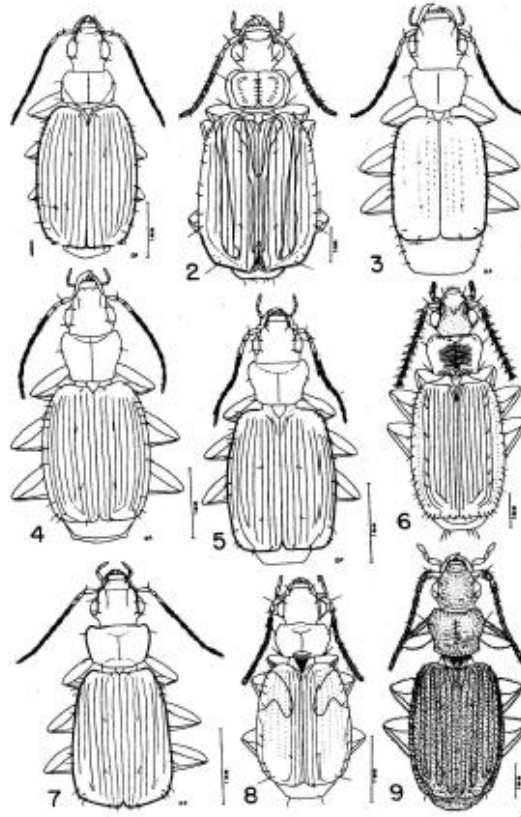
- What are patterns??
 - Are what **deviates from randomness**;
 - **Recur** with some probability;
 - **Are markers** of some object/event of interest;
 - Are “some” **order** in the chaos;
 - Are what **enable machines to recognize**.

Examples of Patterns: Constellations

- Patterns of constellations are represented by 2D planar graphs



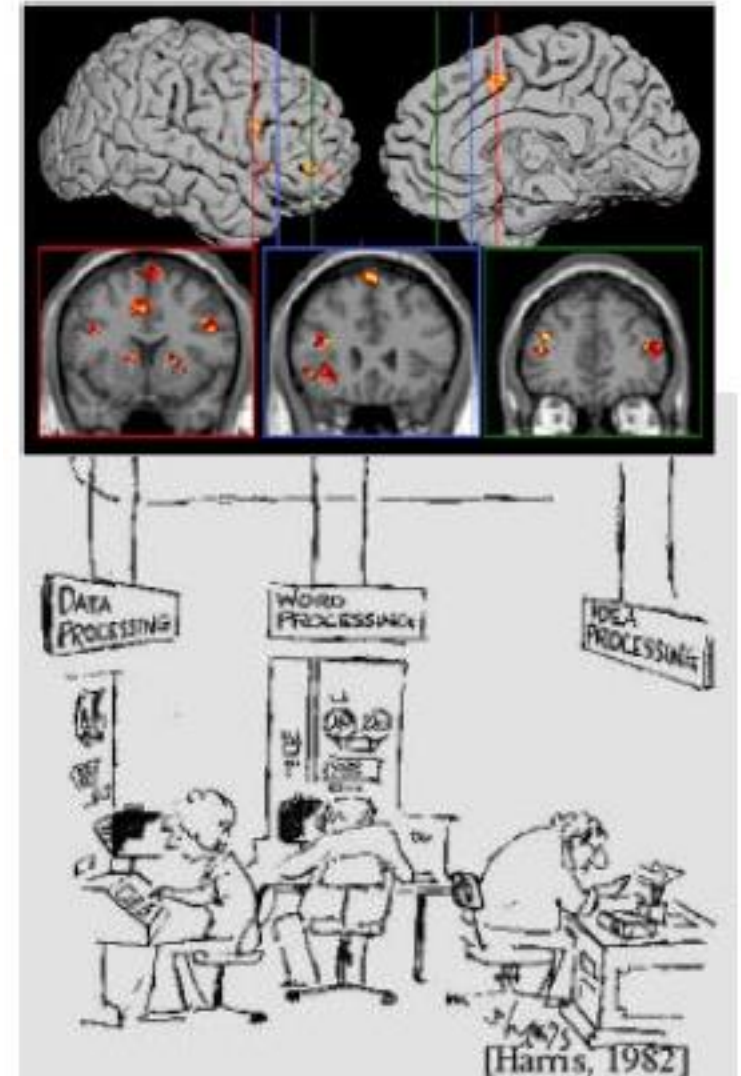
Examples of Patterns: Biological morphology



- Scarabs present different shapes & different colour patterns

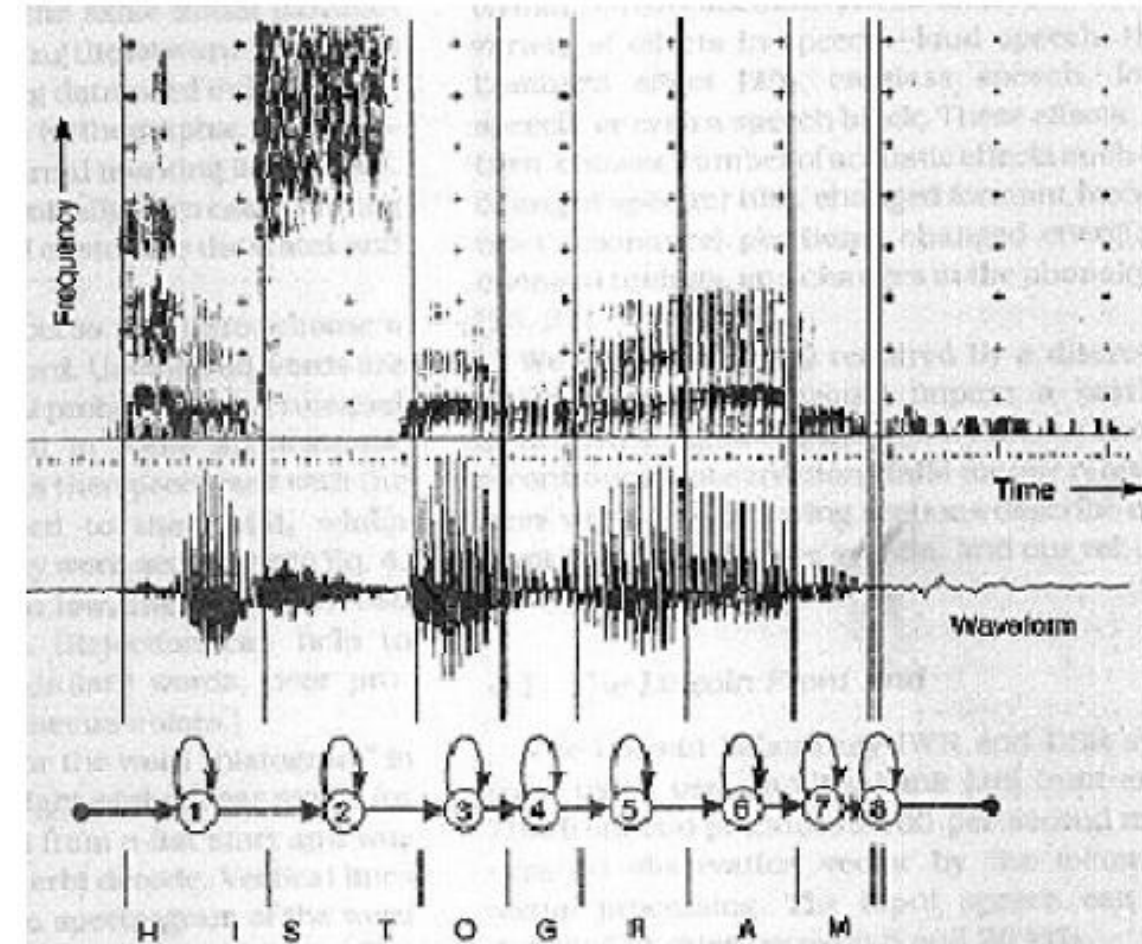
Examples of Patterns: Brain Activity

- Different tasks originate different brain activity patterns
- Find relationships between brain activities, cognition, and behaviours



Examples of Patterns: Speech to Text

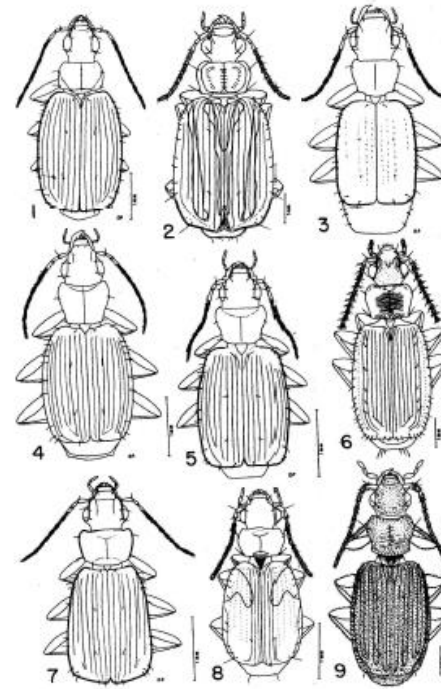
- Different letters/syllables present a different pattern in the time and time-frequency space



ML: Concepts

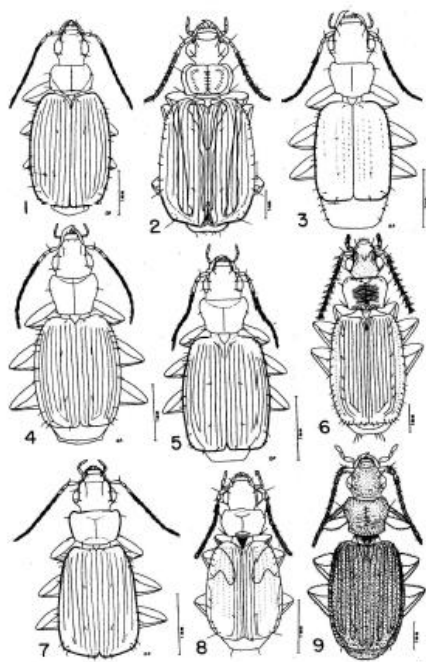
Classification → Assignment of categories (classes or labels) to objects, processes or events

- **Pattern** → “Marker” that enables the labelling, and that emerges from a set of attributes (features)

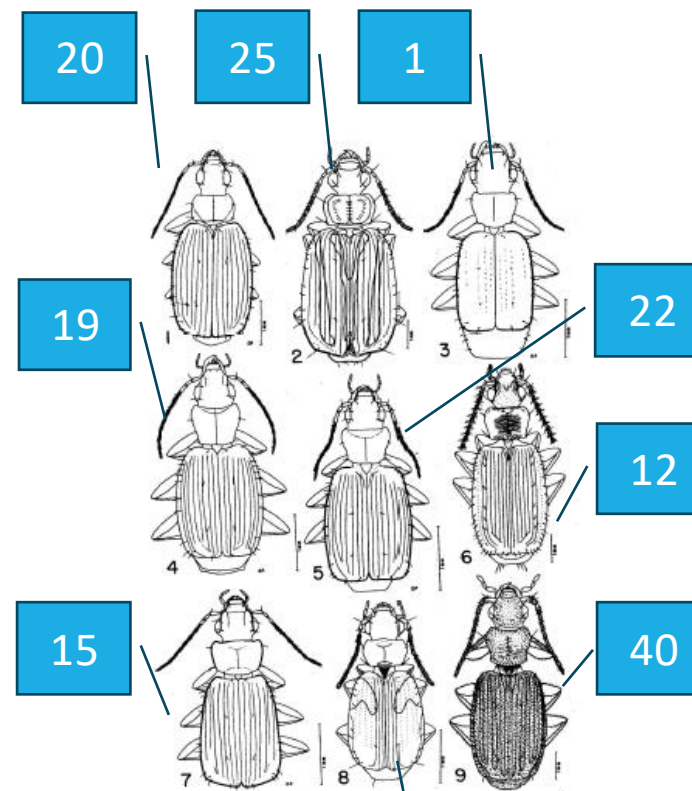


ML: Concepts

- **Features** → Measurements or attributes that represent an object, process or event that enable class assignment (e.g.: CMI in detecting obesity) → **Highlight/identify patterns**

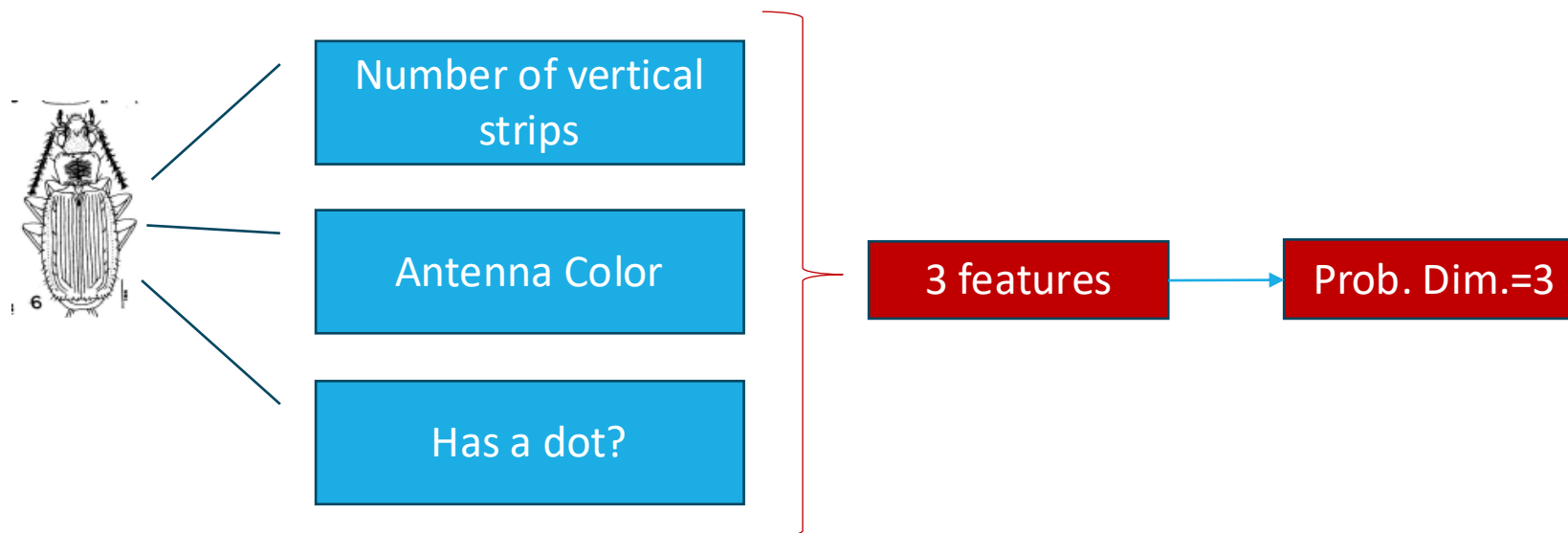


Possible Feature:
Number of Vertical
Strips

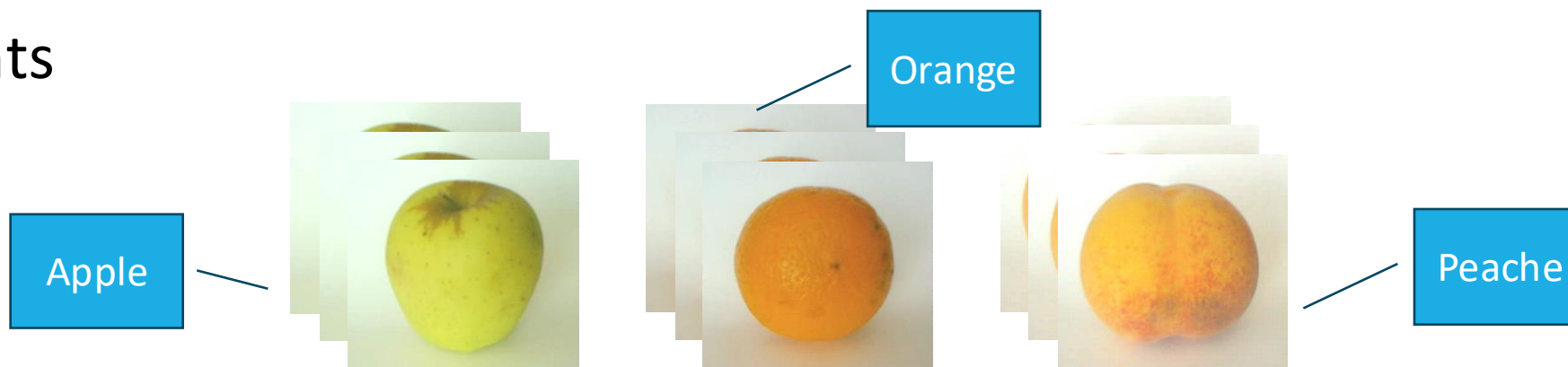


ML: Concepts

- **Problem dimensionality** → Number of features

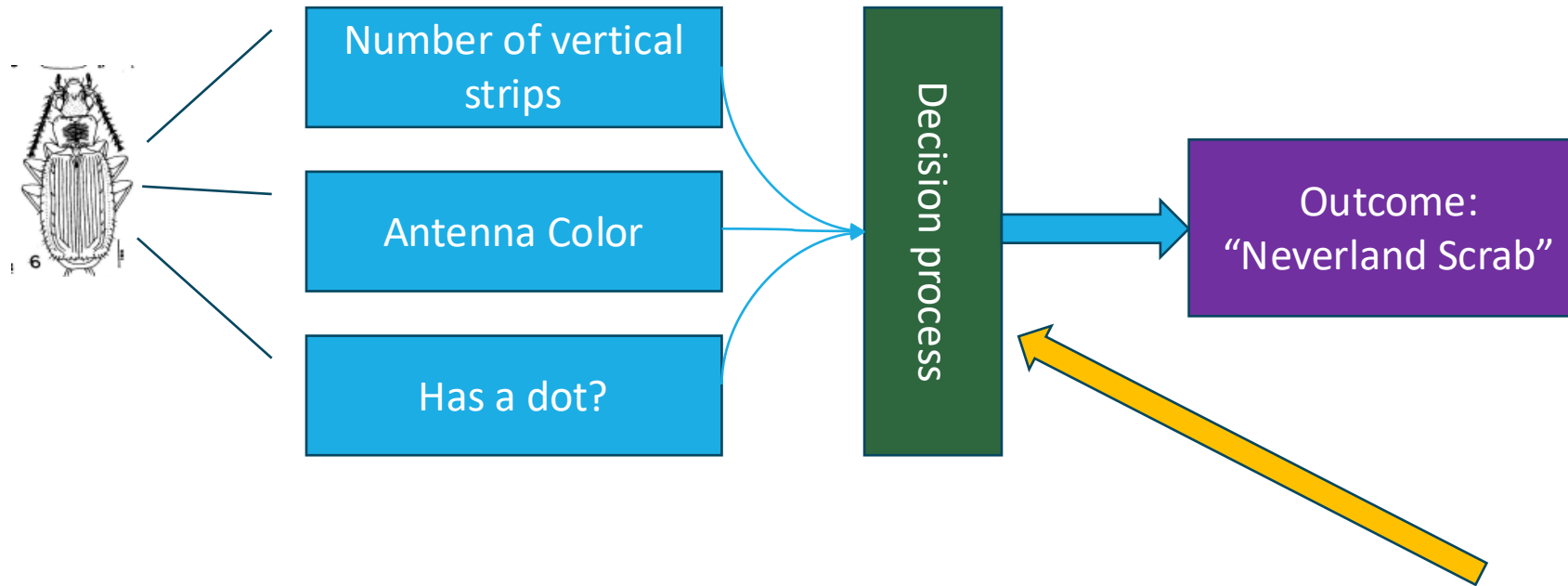


- **Class or Category** → Label assigned to a set of objects, processes or events



ML: Concepts

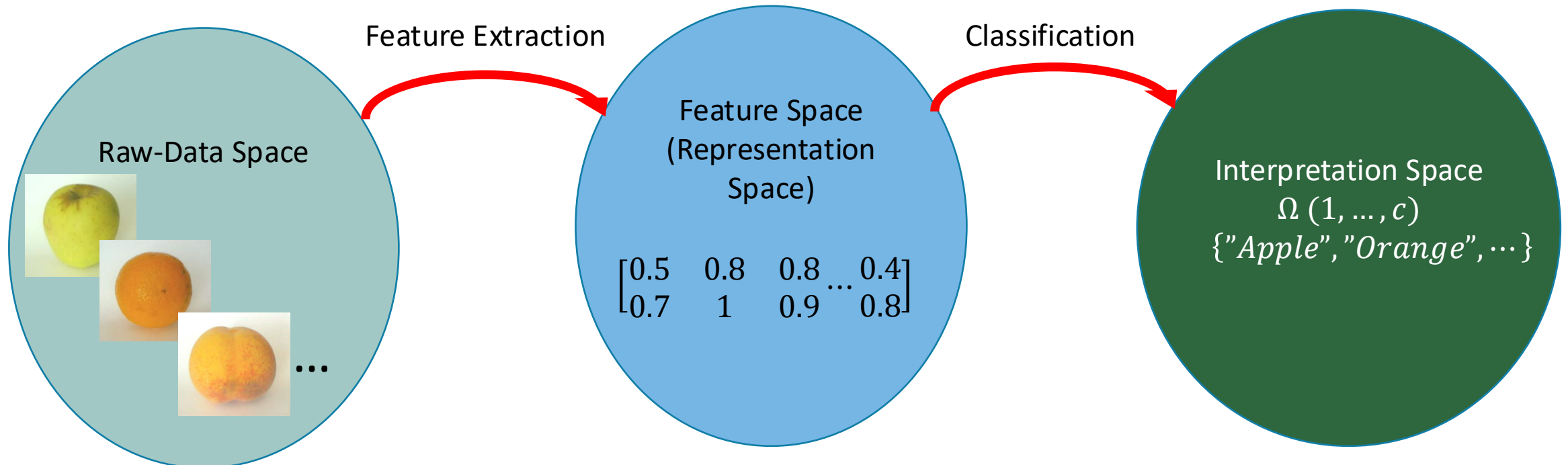
- **Recognition or Classification** → Class assignment



- **Classifier** → Machine that performs class assignment → **Decision process**

ML: Concepts → Spaces

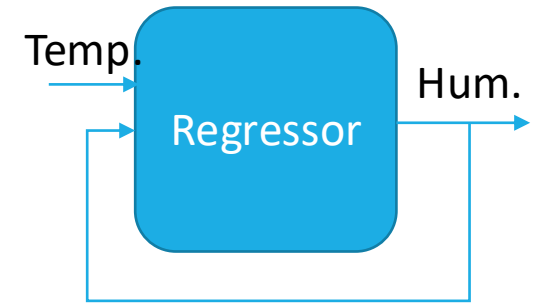
- **Feature Extraction** → Mapping between the raw-data space to the feature space/representation space X
- **Classification** → Mapping between the feature space to the interpretation space Ω
 - → In this course, based in similarity analysis



Main ML: Tasks

- **Classification Task**

- Assign an object to a given class
- Classifiers present discrete output levels



- **Regression Task**

- Regressors infer the continuous value of a given variable based on the past value of the variable (auto-regressive) and/or on the actual and past values of other variables
- A regression task can be converted in a classification task by discretizing the regressor output

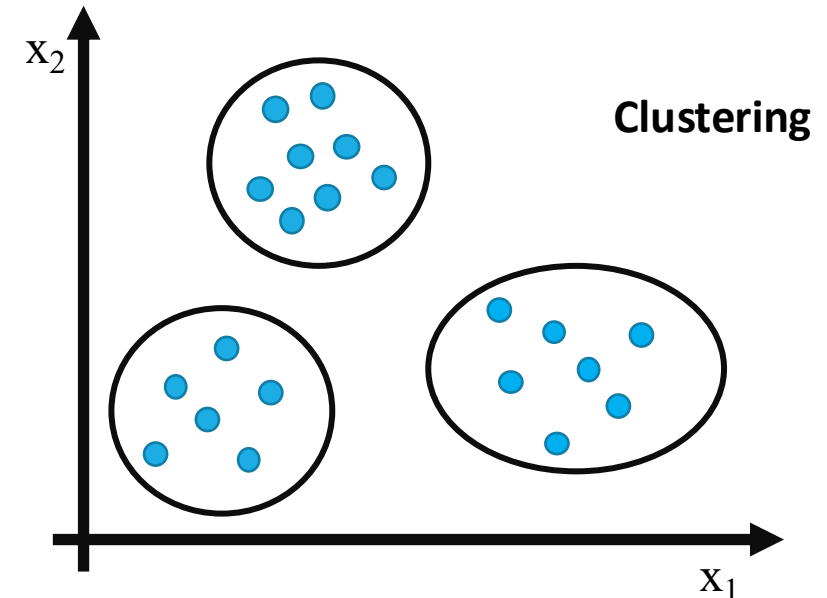
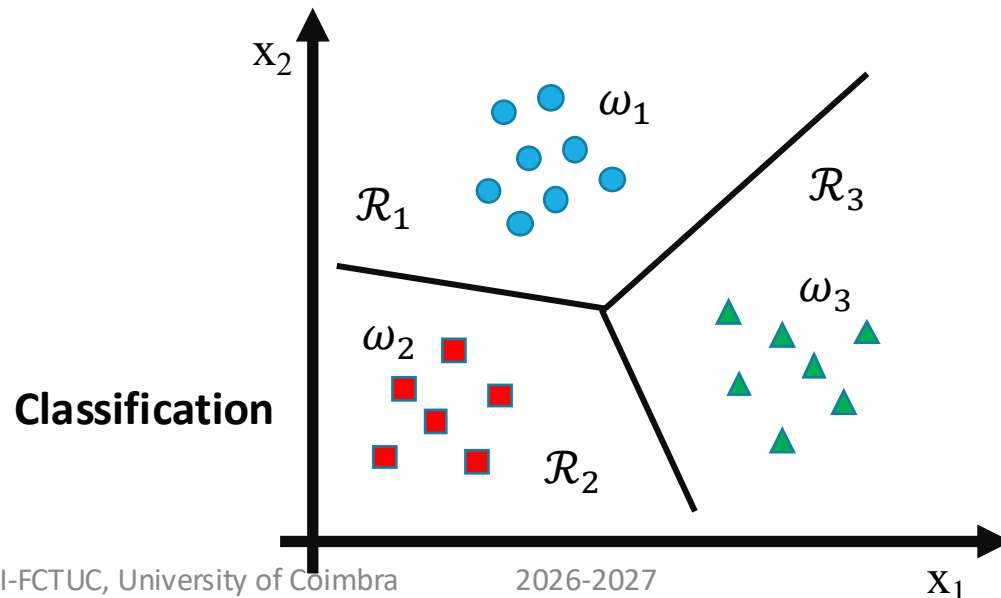
Supervised vs Unsupervised

- **Supervised** → Concept/Prototype driven:
 - Requires the **existence of class labels** (target)
- **Unsupervised** → Driven by inter-element similarity (closeness):
 - Find data structure in the representation space **without knowing nothing about the object labels**
 - Find a useful hypothesis about separability **based only on similarity** relations between objects

ML: Supervised vs Non-Supervised

- Classification vs Clustering

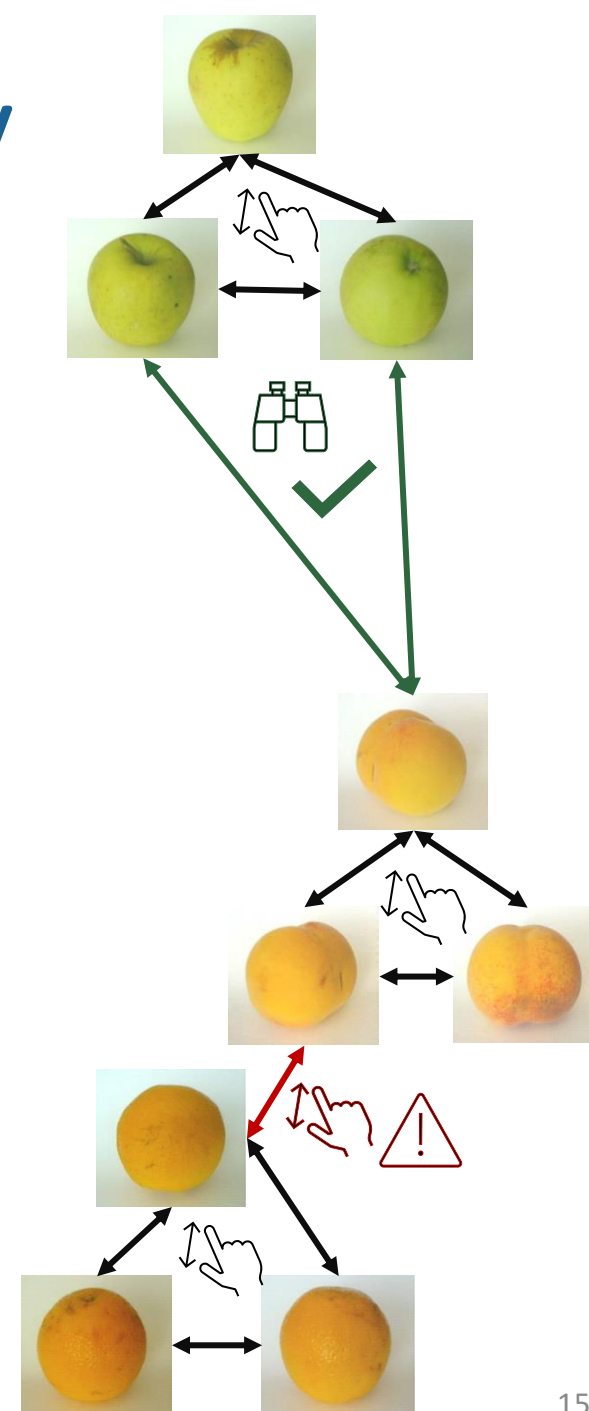
- **Classification** → Supervised Learning → Labels are required to build the PR System
- **Clustering** → Unsupervised Learning → Discovery of categories based just on similarity



Unsupervised Classification: Similarity

- **Similarity**

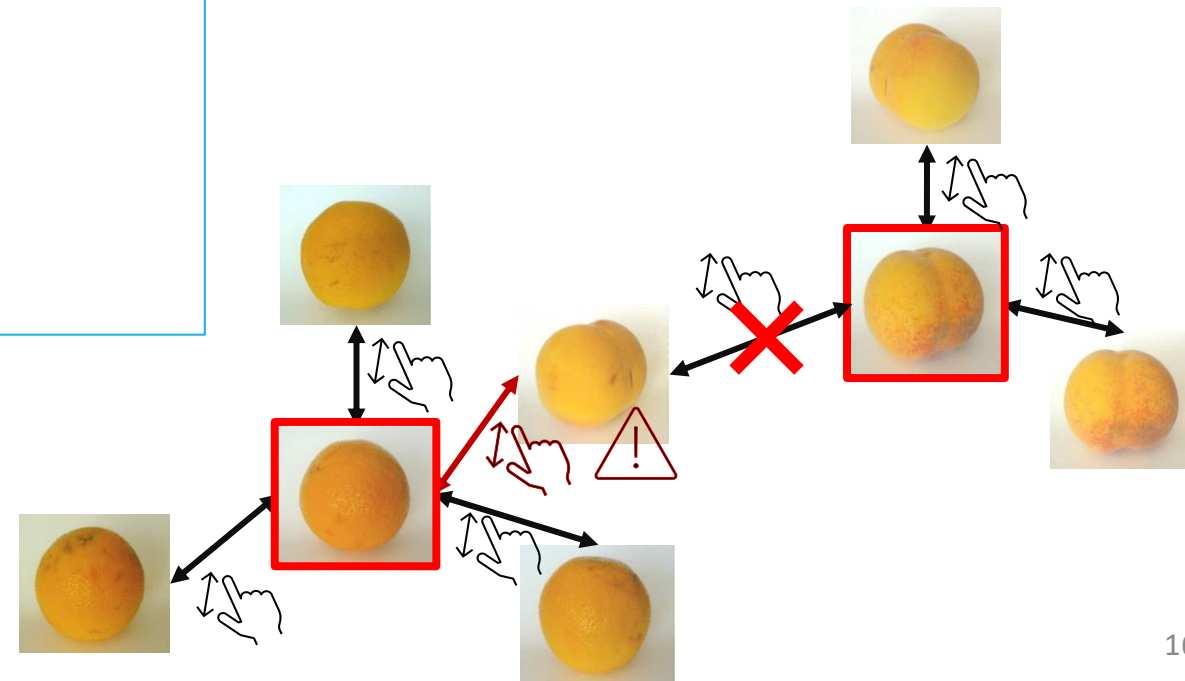
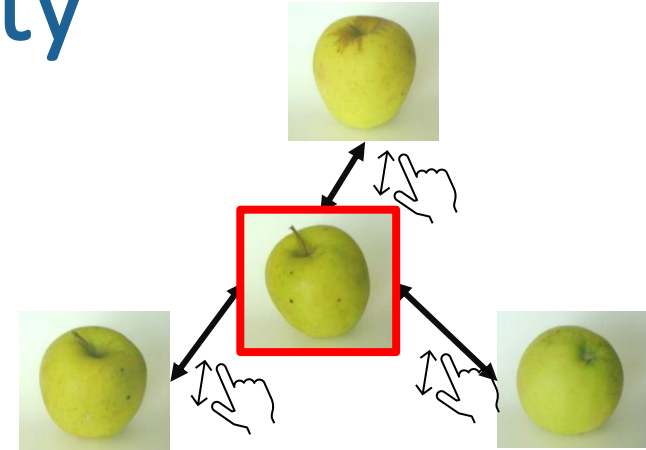
- Two objects are **similar and belong to the same group** if they have similarly valued common attributes (features)
 - **The basis of unsupervised clustering**



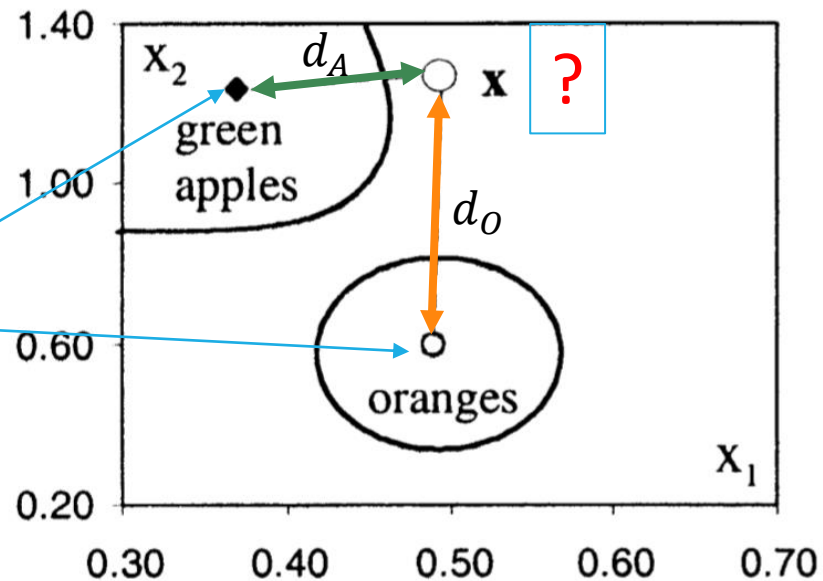
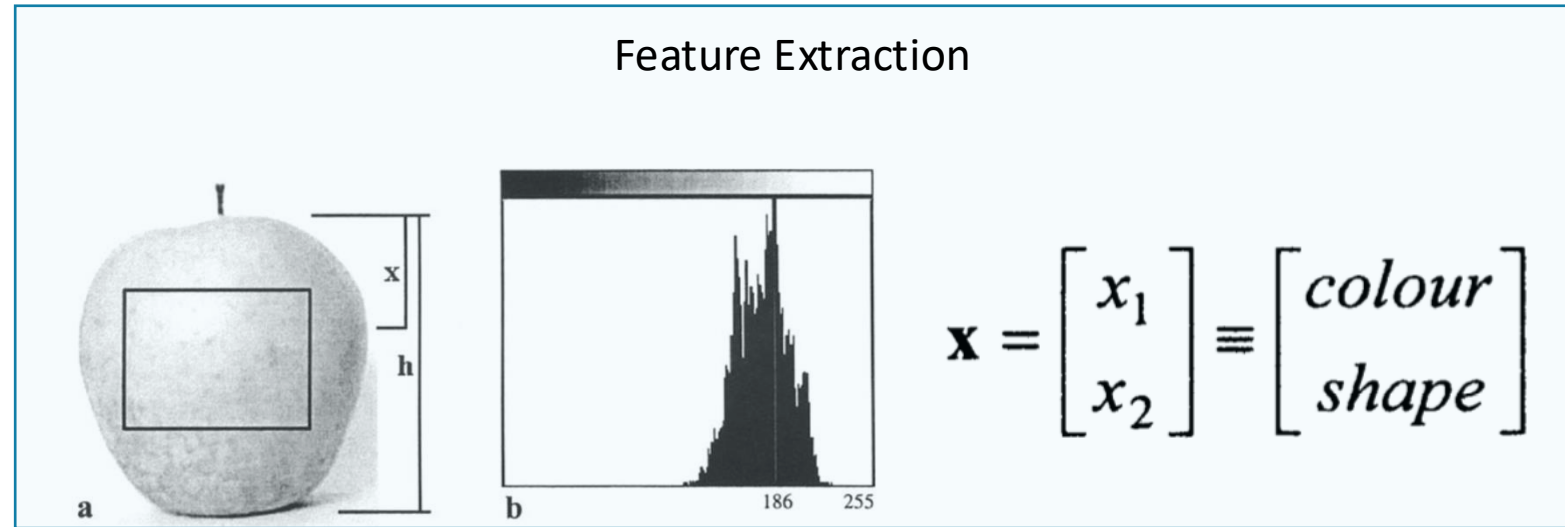
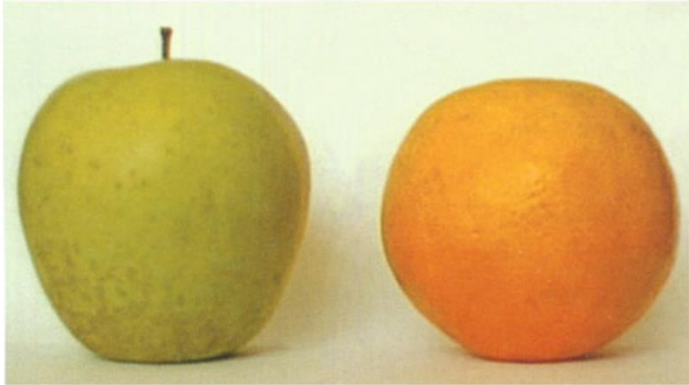
Supervised Classification: Similarity

- **Similarity**

- The basis of pattern recognition
- An object belongs to a given class/category if it presents attributes close to the ideal object, concept, or prototype characteristics.
- In this course, the **distance** between **feature vectors** is used as similarity measures



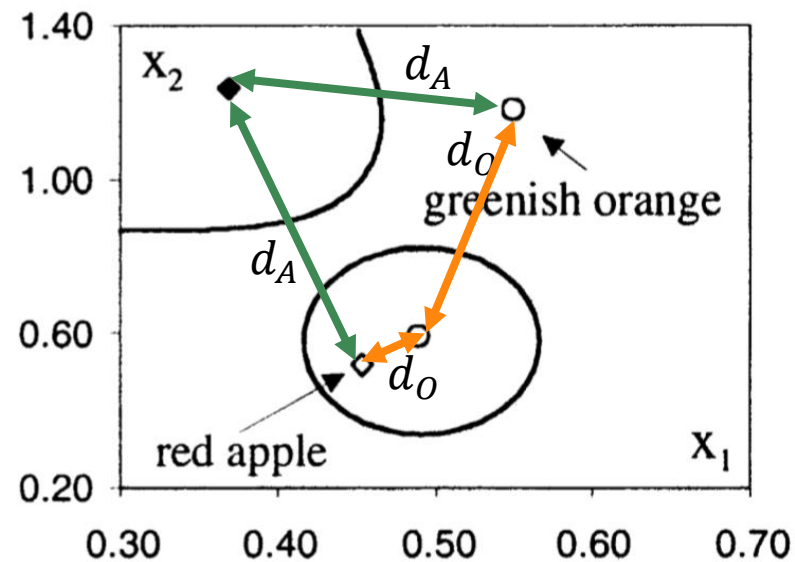
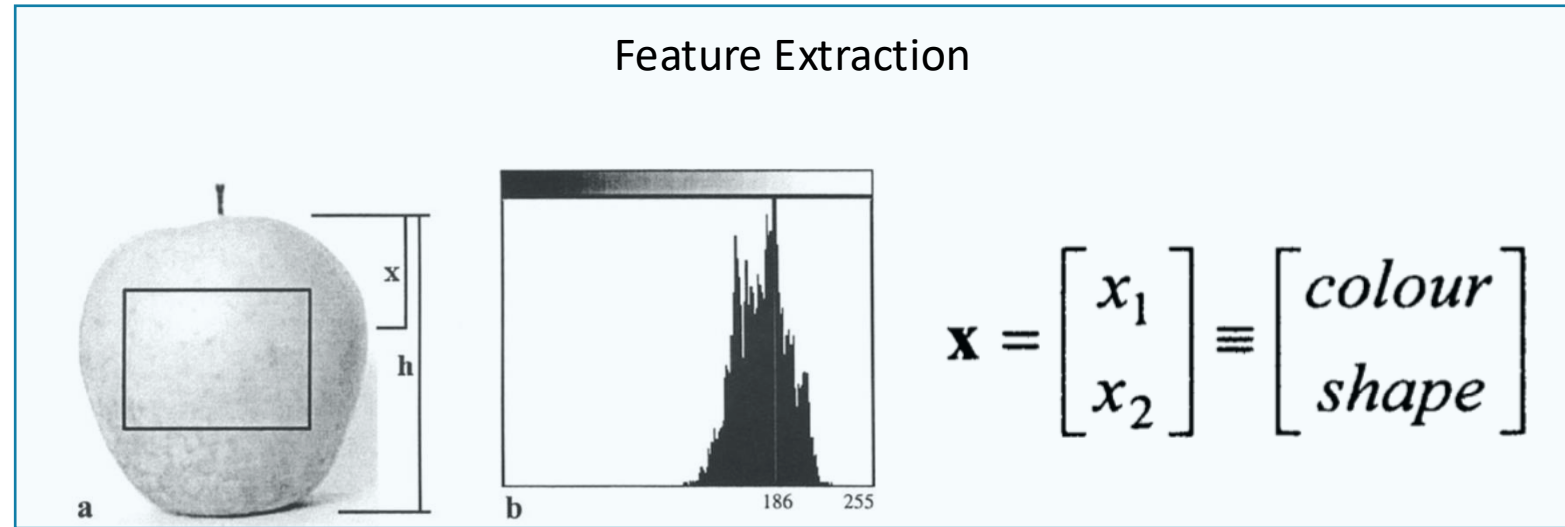
Supervised Classification: Similarity



“Ideal” fruits
(Prototypes)

$$d_A < d_O \Rightarrow x \text{ is "Green Apple"}$$

Supervised Classification: Similarity



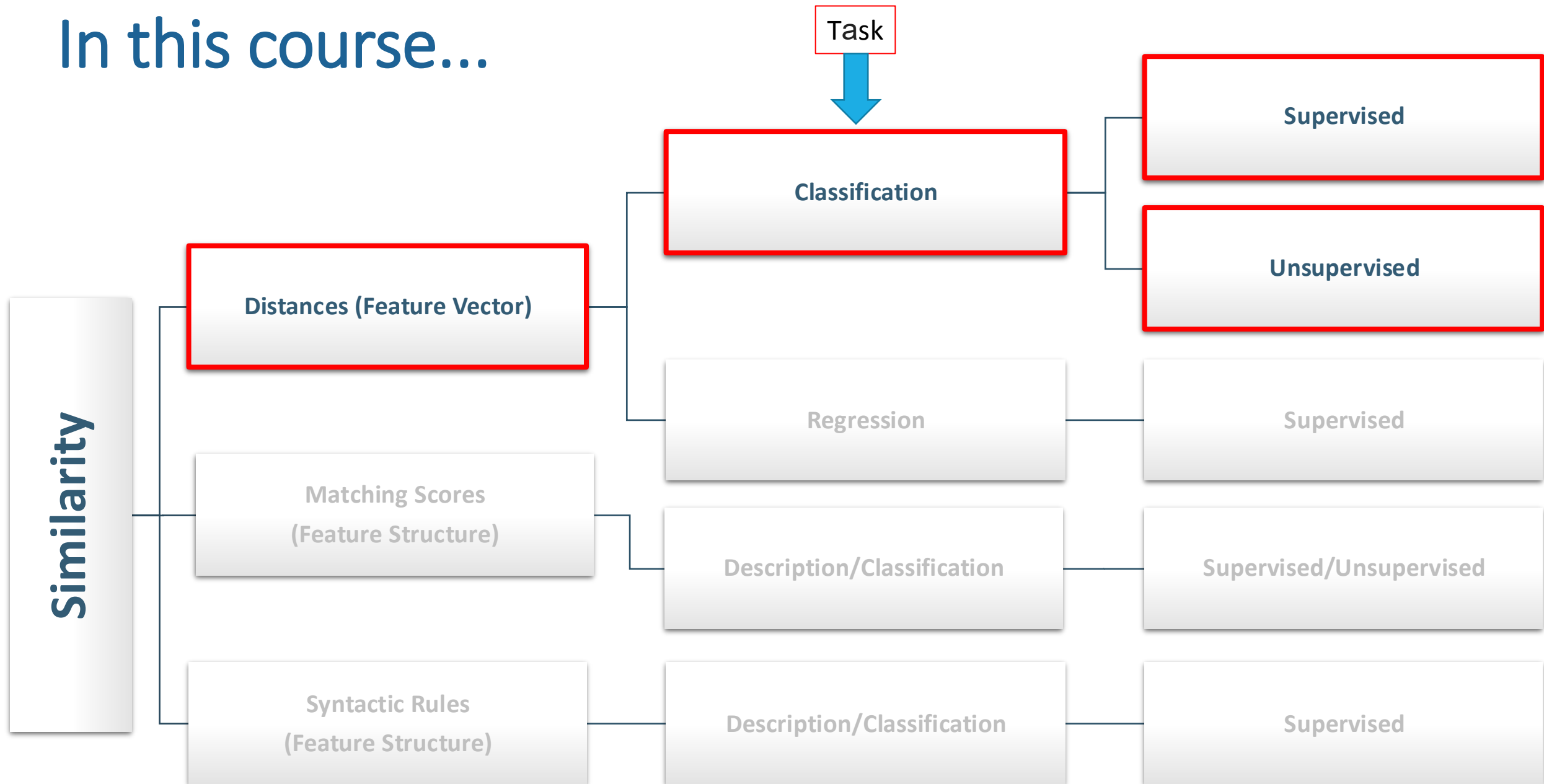
Red Apple

$$d_O < d_A \Rightarrow \text{"Orange"}$$

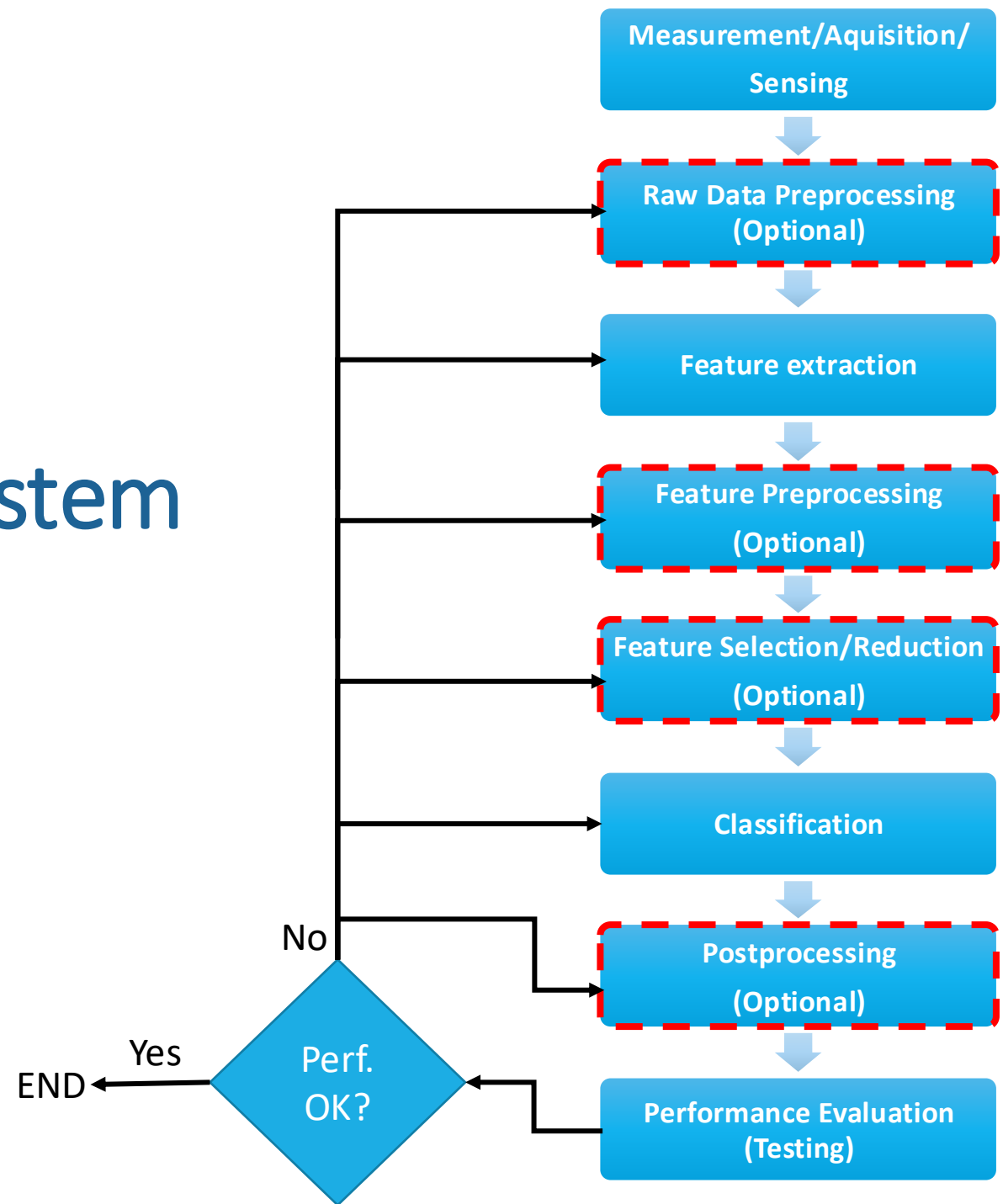
Greenish Orange

$$d_O \approx d_A \Rightarrow \text{"Orange" or "Green Apple" ?}$$

In this course...



Components of a ML System



ML: Some important issues

- **Feature Choice**

- Features should be Discriminative & Not Redundant

- **Classifier parametrization**

- Use many different procedures for training classifiers and choosing models.
- Look for appropriate parametrization (DO NOT TRUST in predefined parameters)

- **Model Evaluation**

- Always use independent datasets for evaluation → Report always performance in testing never in training alone

Examples of Application Areas

- **Biometrics**

- Face recognition, verification, retrieval.
- Fingerprints recognition.
- Speech recognition.

- **Diagnostic systems**

- Medical diagnosis: X-Ray, ECG analysis.
- Machine diagnostics

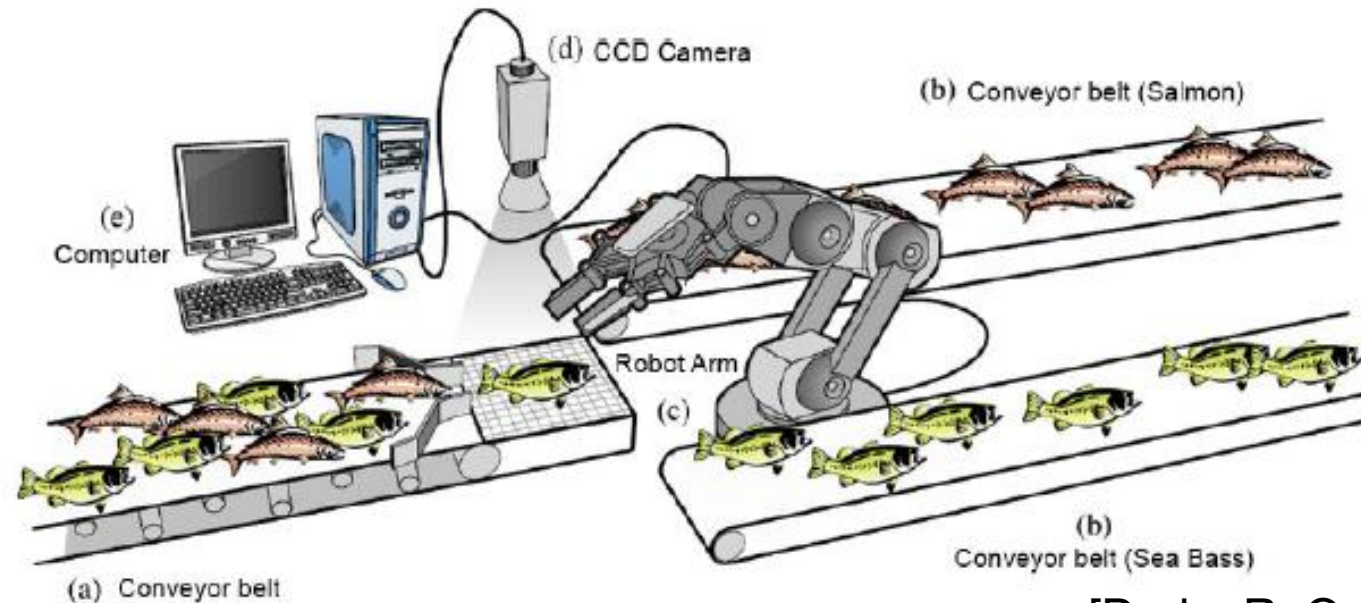
- **Military applications**

- Automated Target Recognition (ATR).
- Image segmentation and analysis (recognition from aerial or satellite photographs).

- **Industry ...**

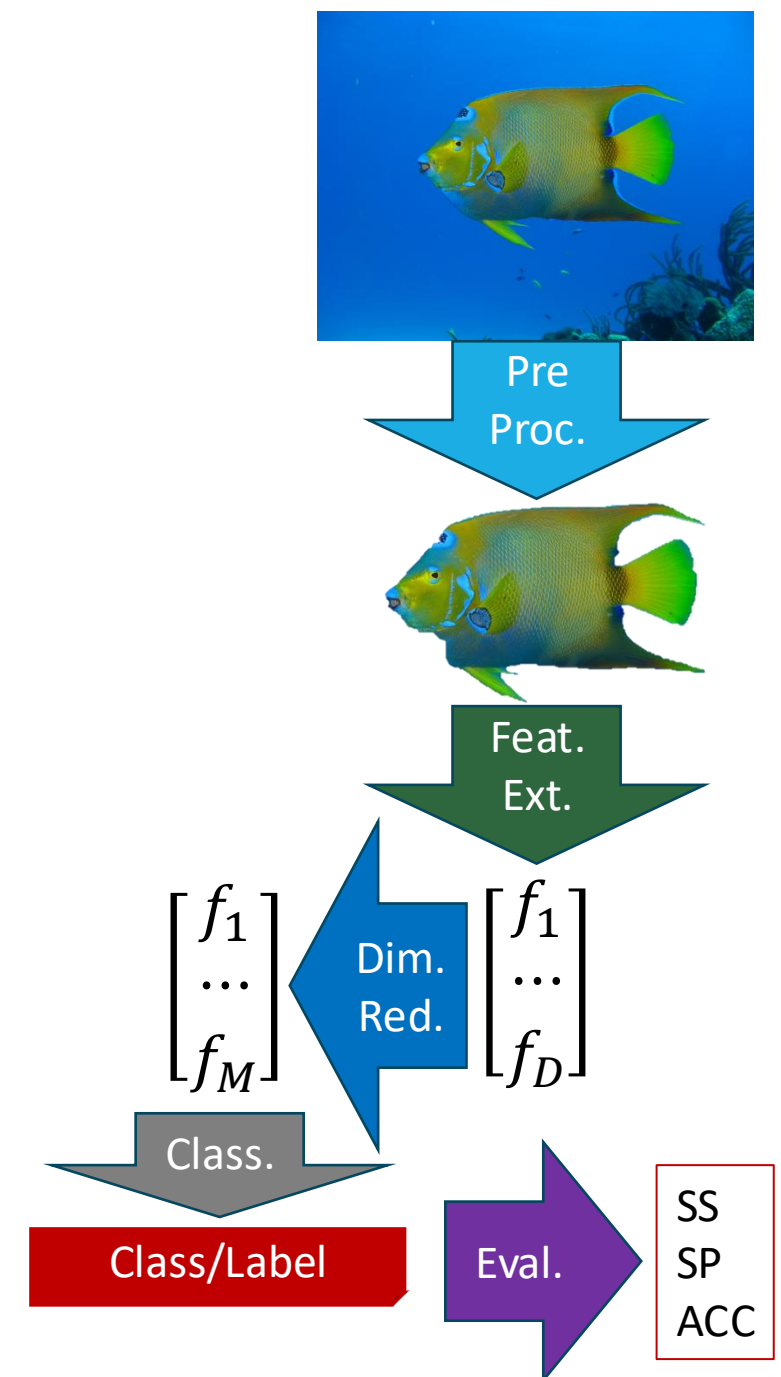
Example (Industry): Automatic Fish Classification System

- Classification of fishes into two classes: Salmon or Sea Bass
 - a) Conveyor belt for all fishes
 - b) Two conveyor belts for classified fishes
 - c) Robotic arm commanded by a computer that separates fishes
 - d) CCD Camera acting as the sensing device
 - e) Computer with a classification system implemented

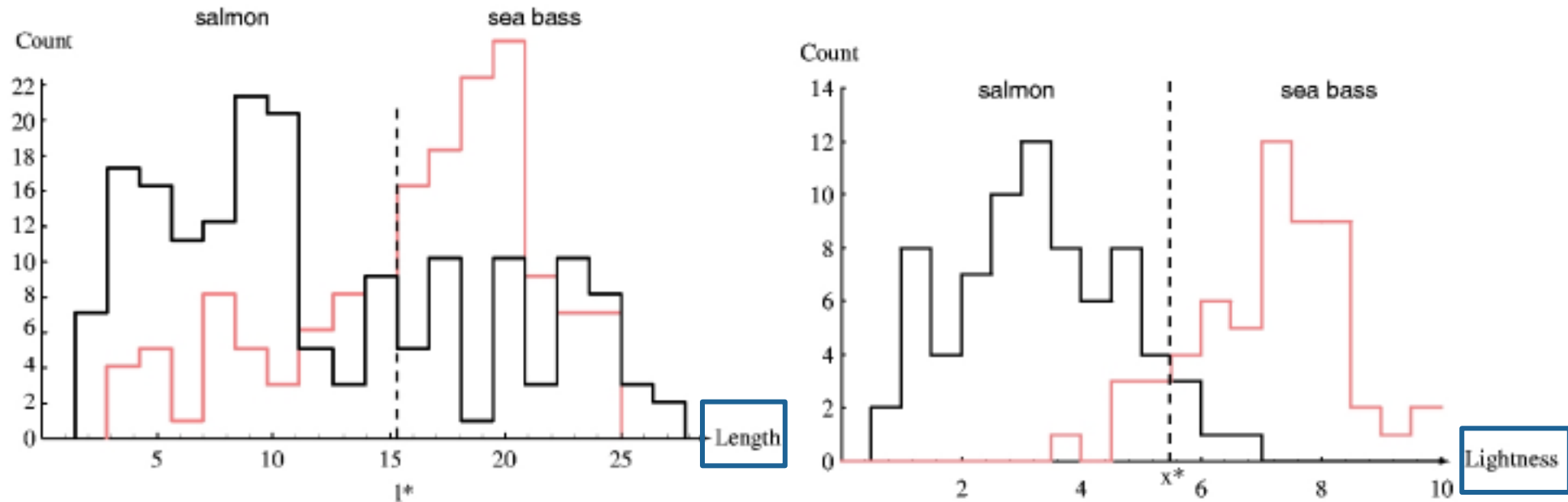


Automatic Fish Classification System

- Preprocessing (e.g. isolate fish from one another and from background with image segmentation)
- Feature Extraction
 - e.g. extract features from each fish (length, width, brightness, etc.)
- Dimensionality reduction: Feature selection/transf.
- Classification
- Model Evaluation



Automatic Fish Classification System: Feature Selection



- The length is a poor feature alone! **Why?**
 - **Because feature values are in great extension superposed for the different classes!!**
- Select the lightness as a possible feature.

- “Customers do not really want sea bass in their cans of salmon, but tolerate salmon in sea bass cans”

- → It is more costly to classify wrongly a sea bass as salmon

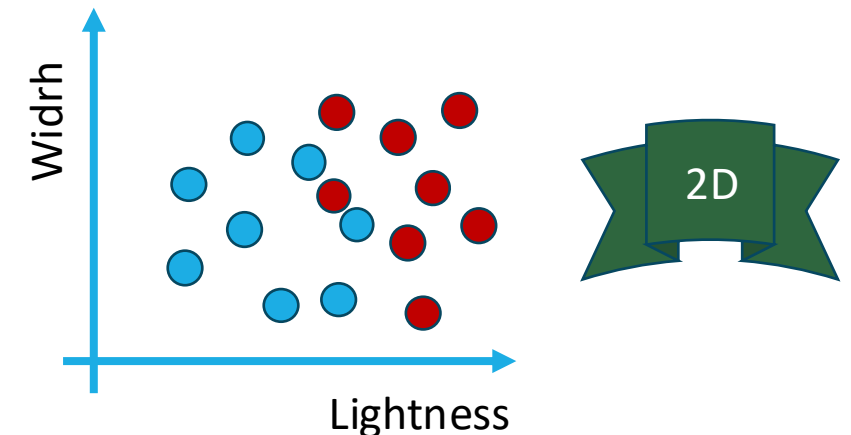
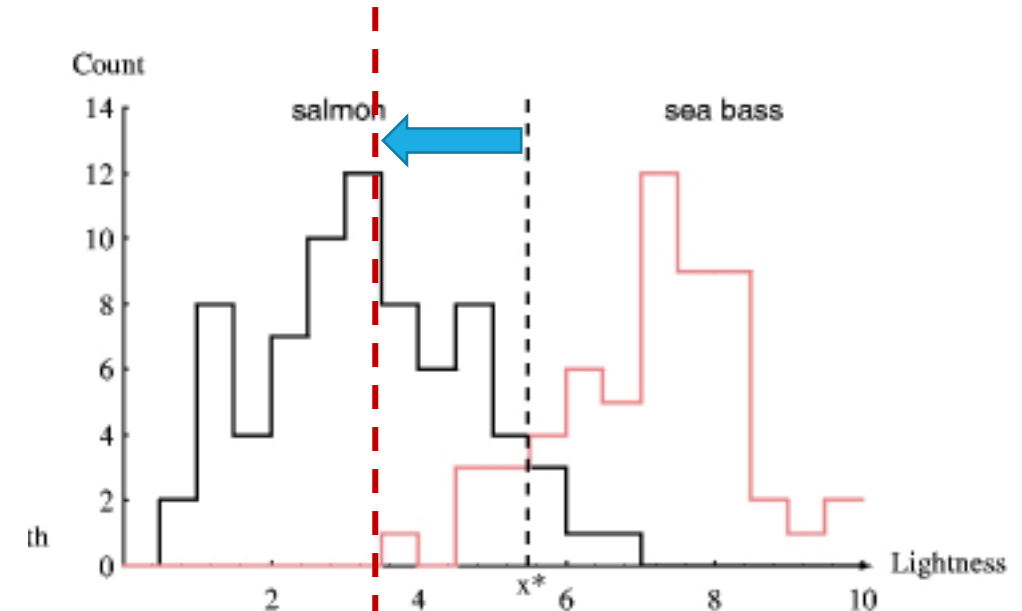
- **How to deal with this??**

- Change decision boundary taking in account cost
- Move our decision boundary toward smaller values of lightness to minimize the cost (reduce the number of sea bass that are wrongly classified as salmon!)

- Other improvement: Adopt the “**lightness**” and add “**width**” of the fish, becoming the feature vector:

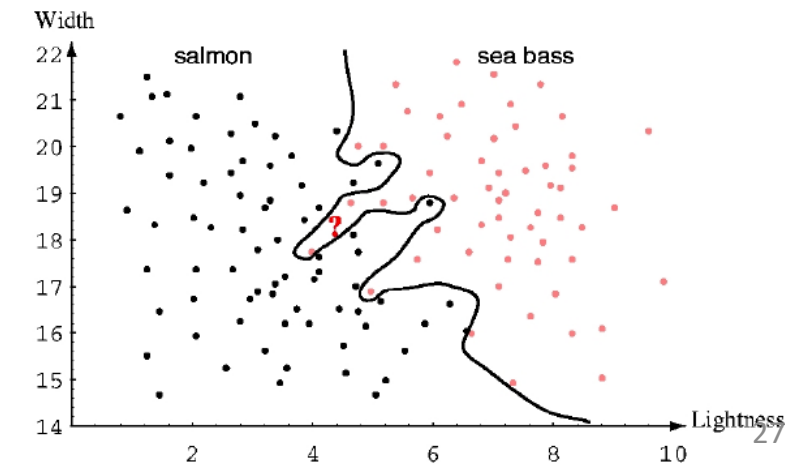
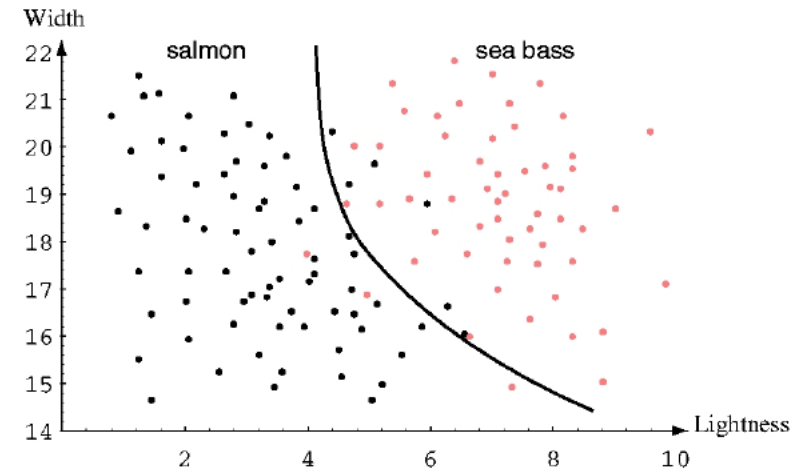
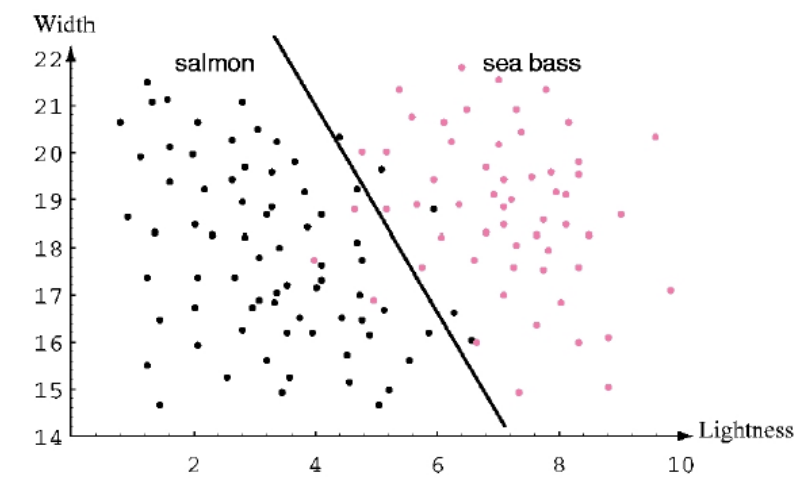
$$Fish = \begin{bmatrix} Lightness \\ Width \end{bmatrix}$$

Automatic Fish Classification System: Domain Constraints



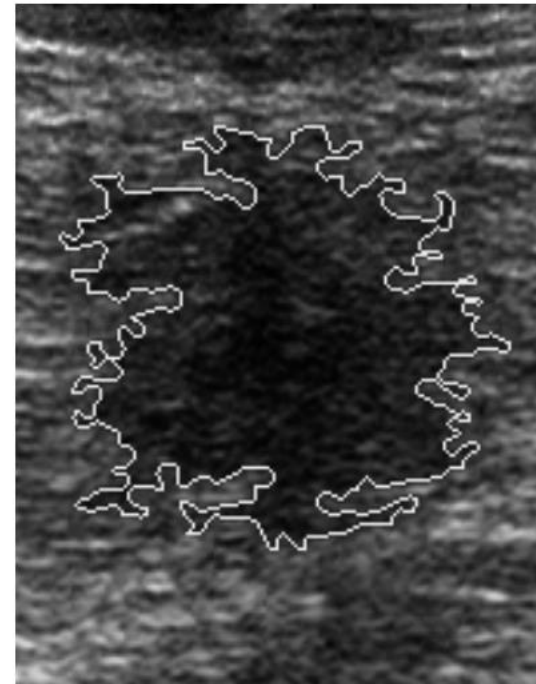
Automatic Fish Classification System: Classification

- Decision boundaries depend on the classifier used
- Ideally, the best decision boundary should be the one which provides an optimal performance
- However, good training performance may not mean good generalization performance. The central aim of designing a classifier is to correctly classify novel inputs → **Issue of generalization!**



Example: Classification of Breast Tumours

- Which procedure you will adopt for feature extraction?
- Which features you will consider?



Malignant

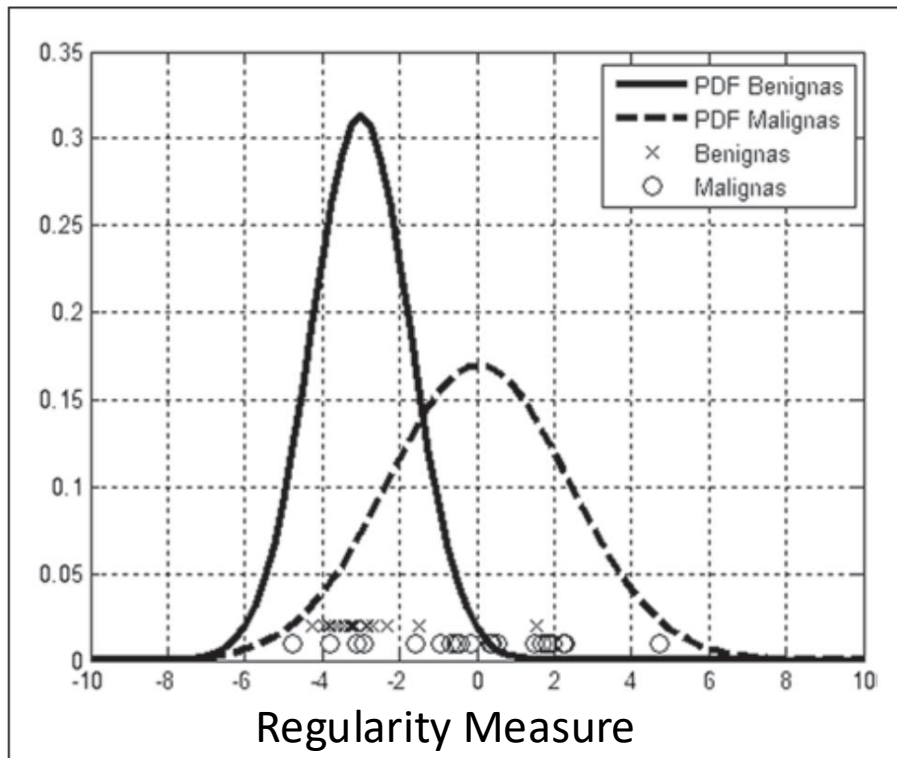
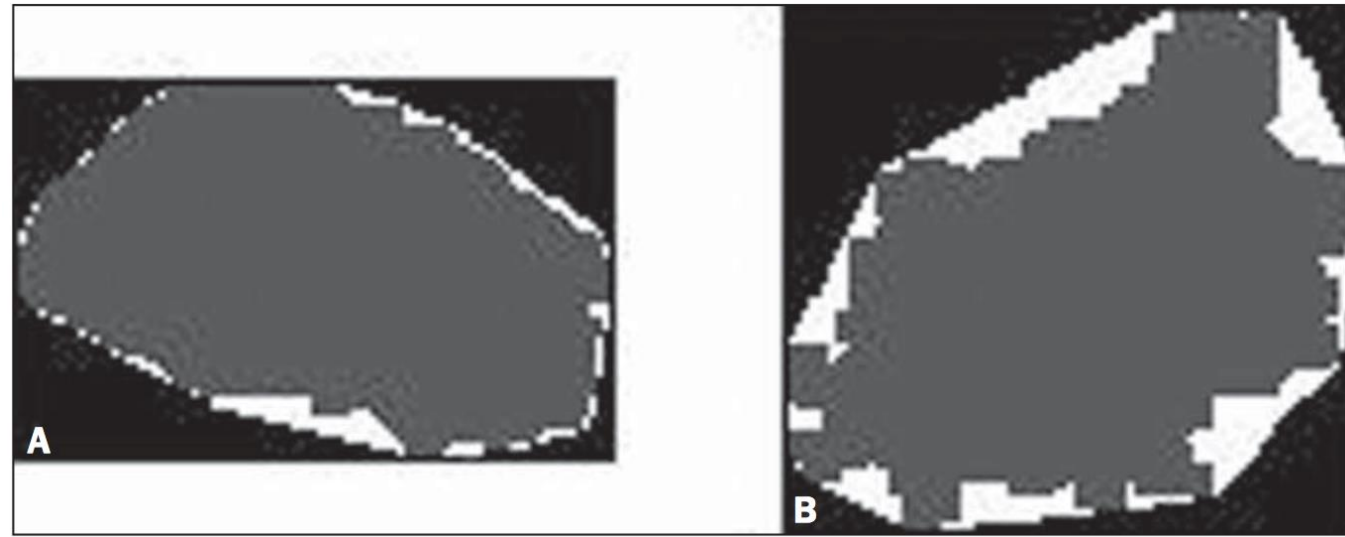


Benign

Alvarenga, A. V., Pereira, W. C., Infantsi, A. F. C., & de Azevedo, C. M. (2005, September). Classification of breast tumours on ultrasound images using morphometric parameters. In *Intelligent Signal Processing, 2005 IEEE International Workshop on* (pp. 206-210). IEEE.

Example: Classification of Breast Tumours

- Features → Measures of irregularity



Calas, M. J. G., Alvarenga, A. V., Gutfilen, B., & de Albuquerque Pereira, W. C. (2011). Avaliação de parâmetros morfométricos calculados a partir do contorno de lesões de mama em ultrassonografias na distinção das categorias do sistema BI-RADS. *Radiologia Brasileira*, 44(5).

Example: Bird song classification

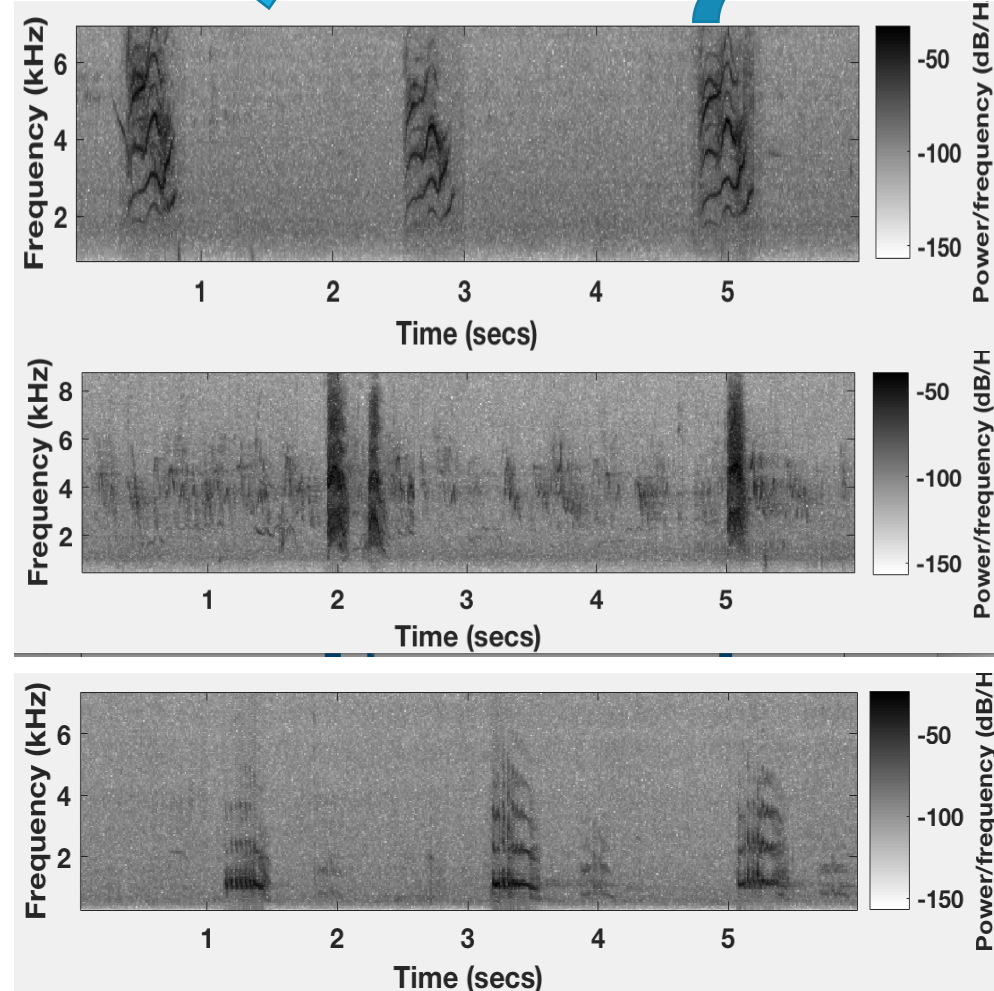
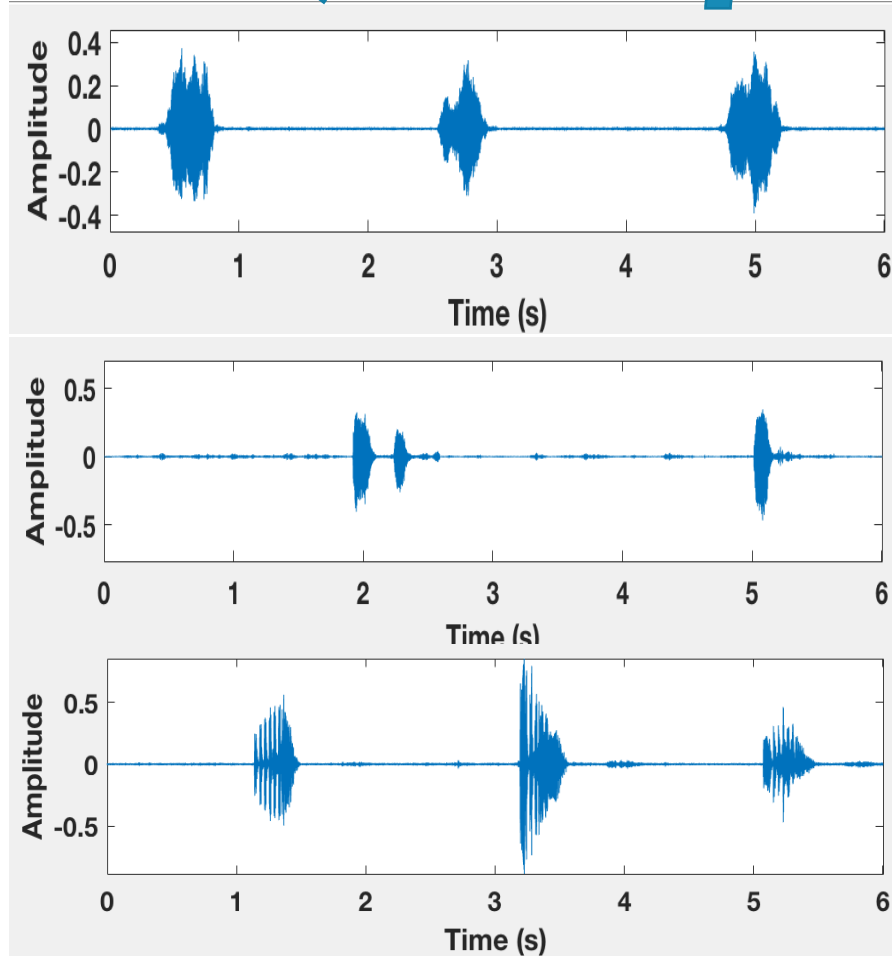


Image
Processing,
Feature
Extraction
&
Classification

<http://www.birdsongs.it/songs/songspectro.asp>

Example: Digit Recognition



- Each Digit is a matrix 16X16!
- What is the number of classes?
 - **10 classes!**
- Features?
 - **Percent of filled pixels above horizontal half point**
 - **Percent of filled pixels to right of vertical half point**
 - **Is reflected y axis**
 - **Is reflected x axis**
 - **...**

<http://archive.ics.uci.edu/ml/datasets/optical+recognition+of+handwritten+digits>

Summary

- Basic Concepts of Pattern Recognition
- Examples of Application Areas
- Pattern Recognition Phases
- Design Cycle of a Pattern Recognition Project